

Charotar University of Science and Technology (CHARUSAT)
Faculty of Technology and Engineering (FTE)

Subject: Web Development Frameworks (ITUE203)

Semester: 3rd

Teaching Scheme:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/Week	3	4	-	7	5
Marks	100	50	-	150	

Course Pre-requisites:

- Overview of Programming Concepts
- Understanding of basic Hypertext Markup Language elements (headings, paragraphs, links, images) and Cascading Style Sheets styling (colors, fonts, and layouts).

Course Description:

This course will cover the basics of the Internet, building a web application, databases, performance and security, and building a web application. In addition, the course will have an extensive set of Practical Tutorials which will help students get a feel for real-world development.

Course Objectives:

1. Equip students with a strong foundation in web development by teaching the core concepts of Hypertext Markup Language, Cascading Style Sheets, and JavaScript to create responsive and interactive web pages.
2. Learn to create robust and scalable back-end systems using PHP, a widely-used server-side scripting language.
3. Integrate front-end and back-end components to build complete, data-driven web applications.
4. Apply best practices for security, including input validation, password hashing, and protection against SQL injection.

Course Outcomes:

By the end of the course, students will be able to:

1. **CO1:** Design and develop responsive and interactive web application using Hypertext Markup Language elements, forms, multimedia, and layouts.
2. **CO2:** Demonstrate Cascading Style Sheets styling techniques such as selectors, text formatting, page formatting, and box model concepts to enhance the presentation of web pages.
3. **CO3:** Implement dynamic behaviours and user interactions using JavaScript (ES6) features, including conditional branching, loops, functions, and operators.
4. **CO4:** Manipulate the Document Object Model to create, update, or remove elements and add interactivity to web applications through event handling.
5. **CO5:** Demonstrate server-side functionality, including handling forms, managing sessions and cookies, and implementing Object-Oriented Programming principles.

➤ **Self-study/Further Study components and materials:**

Unit No.	Unit/Topic Details	
1	INTRODUCTION TO WEB TECHNOLOGIES	
	1.1	Terminologies: Web client, Web server, Web browser, Webpage, Website, Client-server architecture
	Materials: ➤ https://www.geeksforgeeks.org/what-is-a-web-server-working-and-architecture/	
2	BUILDING WEB APPLICATION USER INTERFACE WITH HYPERTEXT MARKUP LANGUAGE 5 AND CASCADING STYLE SHEETS 3	

	2.1	Hypertext Markup Language Introduction
	2.2	Hypertext Markup Language Documents
	Materials: ➤ https://www.w3schools.com/html/html_basic.asp	
	2.3	Cascading Style Sheets Introduction
	Materials: ➤ https://www.w3schools.com/css/css_intro.asp	
3	BUILD INTERACTIVE USER INTERFACE WITH JAVASCRIPT FUNDAMENTALS (ES6)	
	3.1	An Introduction to JavaScript
	3.2	Code editors
	3.3	Developer console
	3.4	Variable and Data Types
	3.5	Conditional branching and loops and iteration
	Materials: ➤ https://javascript.info/intro ➤ https://javascript.info/code-editors ➤ https://javascript.info/devtools ➤ https://javascript.info/variables ➤ https://javascript.info/types ➤ https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide/Loops_and_iteration	
	JAVASCRIPT OBJECTS	
4	4.1	Objects
	Materials: ➤ https://javascript.info/object	
5	BUILD WEB APPLICATION WITH SERVER-SIDE LANGUAGE - PHP	
	5.1	Introduction to PHP, Variables & Data Types
	5.2	Decisions and loops Structure
	Materials: ➤ https://www.tutorialspoint.com/php/php_introduction.htm ➤ https://www.tutorialspoint.com/php/php_variable_types.htm ➤ https://www.tutorialspoint.com/php/php_data_types.htm ➤ https://www.tutorialspoint.com/php/php_strings.htm ➤ https://www.tutorialspoint.com/php/php_loop_types.htm	

Syllabus:

Unit No.	Unit/Topic Details		Hours (hr)	Evaluation Weightage (%)
1	INTRODUCTION TO WEB TECHNOLOGIES		03	6
	1.1	Web Protocols, secure connections, application and development tools		
	1.2	Web site design principles, Planning the site and navigation		
	1.3	Web Application Architecture and it's type, choose the right web application type & architecture		
	1.4	Global server and local server		
2	BUILDING USER INTERFACE OF WEB APPLICATION		06	14
	2.1	User-centered design approach		
	2.2	Color theory and psychology		
	2.3	Typography fundamentals (e.g. Elements, Text, and links)		
	2.4	Layout and composition techniques (Tables and layouts, Lists)		
	2.5	Images and iframe, Entities, Forms		

	2.6	Multimedia, and graphics on a webpage		
	2.7	Properties and its value, Selectors, Text formatting		
	2.8	Page formatting (background, absolute, relative, and fixed layout)		
	2.9	Box Model		
3	BUILD INTERACTIVE USER INTERFACE OF WEB APPLICATION		4.5	10
	3.1	Interaction: alert, prompt, confirm		
	3.2	Type Conversions, operators, maths, Comparisons		
	3.4	Functions, Function expressions, Arrow functions		
4	DOCUMENT OBJECT MODEL (DOM)		4.5	10
	4.1	Introduction of Document Object Model, Document Object Model over the JavaScript		
	4.2	Access document element and style		
	4.3	Event listener operations and event handler to an element		
5	JAVASCRIPT OBJECTS		7.5	16
	5.1	Object references, and copying, Garbage collection		
	5.2	Object methods, "this", Constructor, operator "new"		
	5.3	Object to primitive conversion, Methods of primitives		
	5.4	Prescriptive Analytics - Optimization Techniques, Simulation Modelling		
	5.5	Numbers, Strings, Arrays, Array methods		
6	BUILD WEB APPLICATION WITH SERVER-SIDE LANGUAGE		7.5	16
	6.1	Function		
	6.2	Array		
	6.3	Handling Form with user interface		
	6.4	Session and Cookie		
	6.5	Working with file and Directories		
	6.6	Object Oriented Programming (OOPs) Concept: Classes, Object, Properties, Constructor and destructor		
	6.7	Advance Object-Oriented Programming (OOPs) Concept: Interface, Abstract Class, Object cloning		
7	CONNECT WEB APPLICATION DATABASE		06	14
	7.1	Overview of Database		
	7.2	Types of Databases: Relational, NoSQL, Key-Value Stores, Document-based		
	7.3	Understanding Tables, Rows, Columns, and Relationships		
	7.4	Data Manipulation and Data Definition Queries		
	7.5	Where Clause, JOINS, Clone Tables, Temporary Tables		
8	WEB APPLICATION SECURITY		06	14
	8.1	Authentication with Hypertext Transfer Protocol		
	8.2	Understanding security, and some best practices for web apps		
	8.3	Authentication and sessions		
Total Hours:			45	

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	2	3	2	3	-	-	-	1	3	-	-	-	-
CO2	3	2	3	2	3	-	-	-	1	3	-	-	-	-
CO3	3	3	3	2	3	-	-	-	1	3	-	-	-	-
CO4	3	3	3	2	3	-	-	-	1	3	-	-	-	-
CO5	3	3	3	3	3	-	-	-	1	3	2	3	-	-

Recommended coding Languages and tools:

1. Hypertext Markup Language 5 (HML5)
2. Cascading Style Sheets 3(CSS3)
3. MySQL and MongoDB

Recommended Study Material:

❖ **Text books:**

1. "HTML and CSS: Design and Build Websites", (1st Edition) by Jon Duckett.
2. "JavaScript and jQuery: Interactive Front-End Web Development", (1st Edition) by Jon Duckett.
3. "Eloquent JavaScript: A Modern Introduction to Programming", (3rd Edition) by Marijn Haverbeke.
4. "JavaScript: The Definitive Guide", (7th Edition) by David Flanagan.

❖ **Reference books:**

1. "Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics", (5th Edition) by Jennifer Robbins.
2. "You Don't Know JS", Series (6 books) by Kyle Simpson.
3. "Modern JavaScript for the Impatient", by Cay S. Horstmann.
4. "DOM Scripting: Web Design with JavaScript and the Document Object Model", (1st Edition) Jeremy Keith.
5. "Web Development with Node and Express: Leveraging the JavaScript Stack", (6th Edition) by Ethan Brown.

❖ **Additional Online Resources:**

1. MDN Web Docs (Mozilla) URL:
 - **HTML:** <https://developer.mozilla.org/en-US/docs/Web/HTML>
 - **CSS:** <https://developer.mozilla.org/en-US/docs/Web/CSS>
 - **Javascript:** <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
2. JavaScript Info URL:
 - **Javascript:** <https://javascript.info/>

❖ **Online Courses**

1. IBM Full Stack Software Developer
Link: <https://www.coursera.org/professional-certificates/ibm-full-stack-cloud-developer>
2. Web Development Fundamentals (IBM Skillsbuild)
Link: <https://students.yourlearning.ibm.com/activity/PLAN>

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*MTc0MjE4NzkyMS4xLjAuMTc0MjE4NzkyMS4wLjAuMA..&utm_source=skillsbuild.org

3. HTML, CSS, and Javascript for Web Developers Specialization

Link: <https://www.coursera.org/specializations/html-css-javascript-for-web-developers>